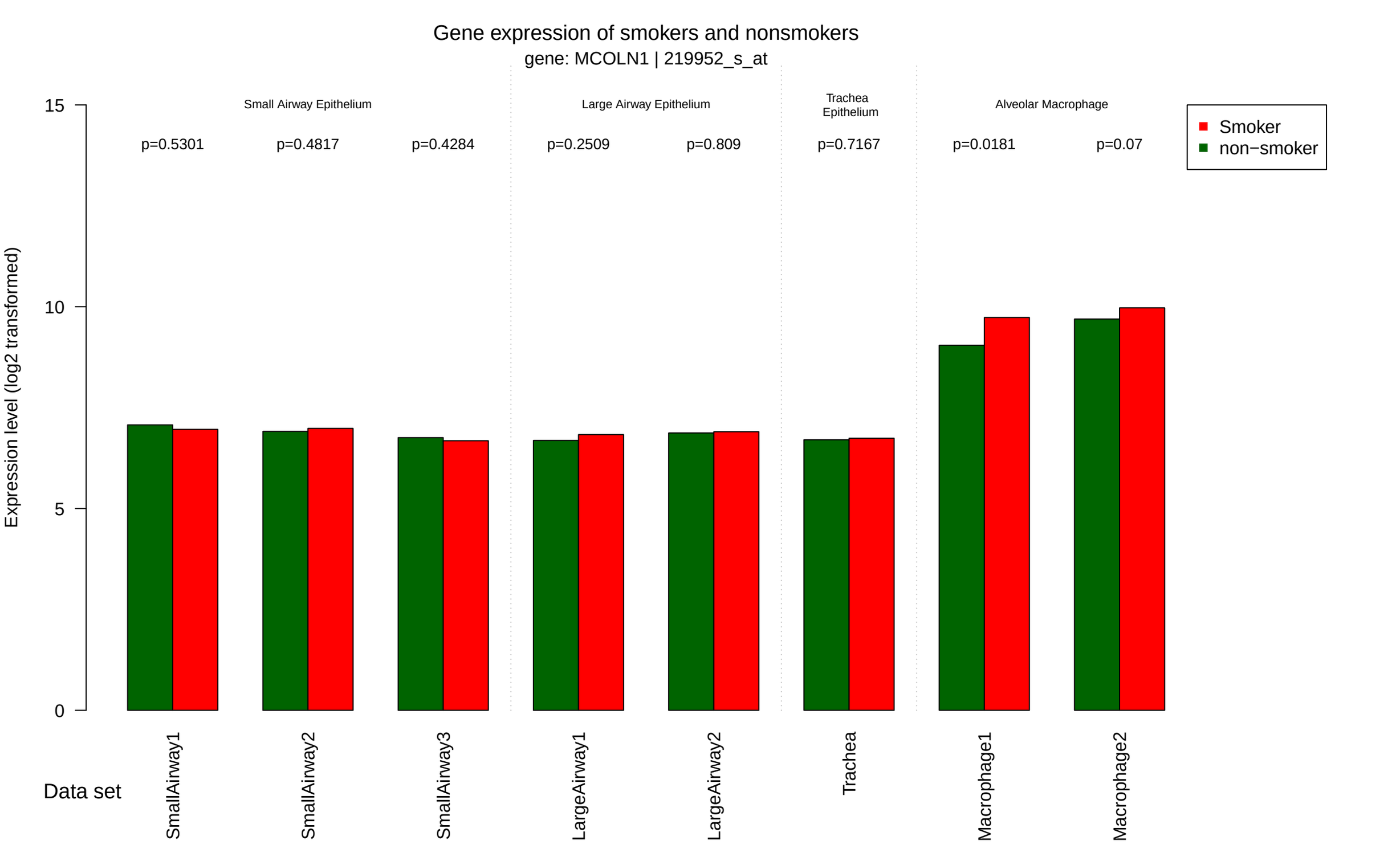
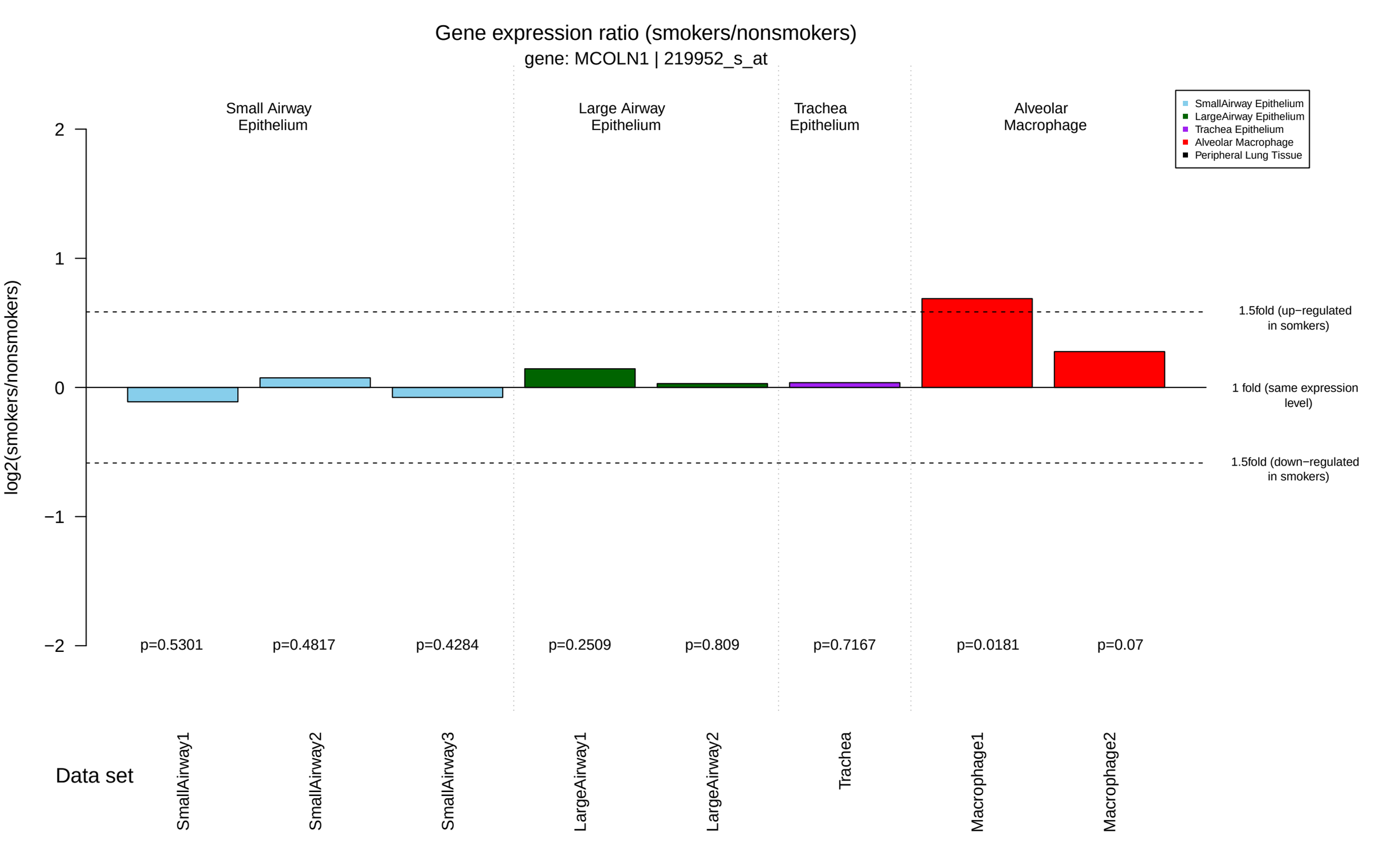
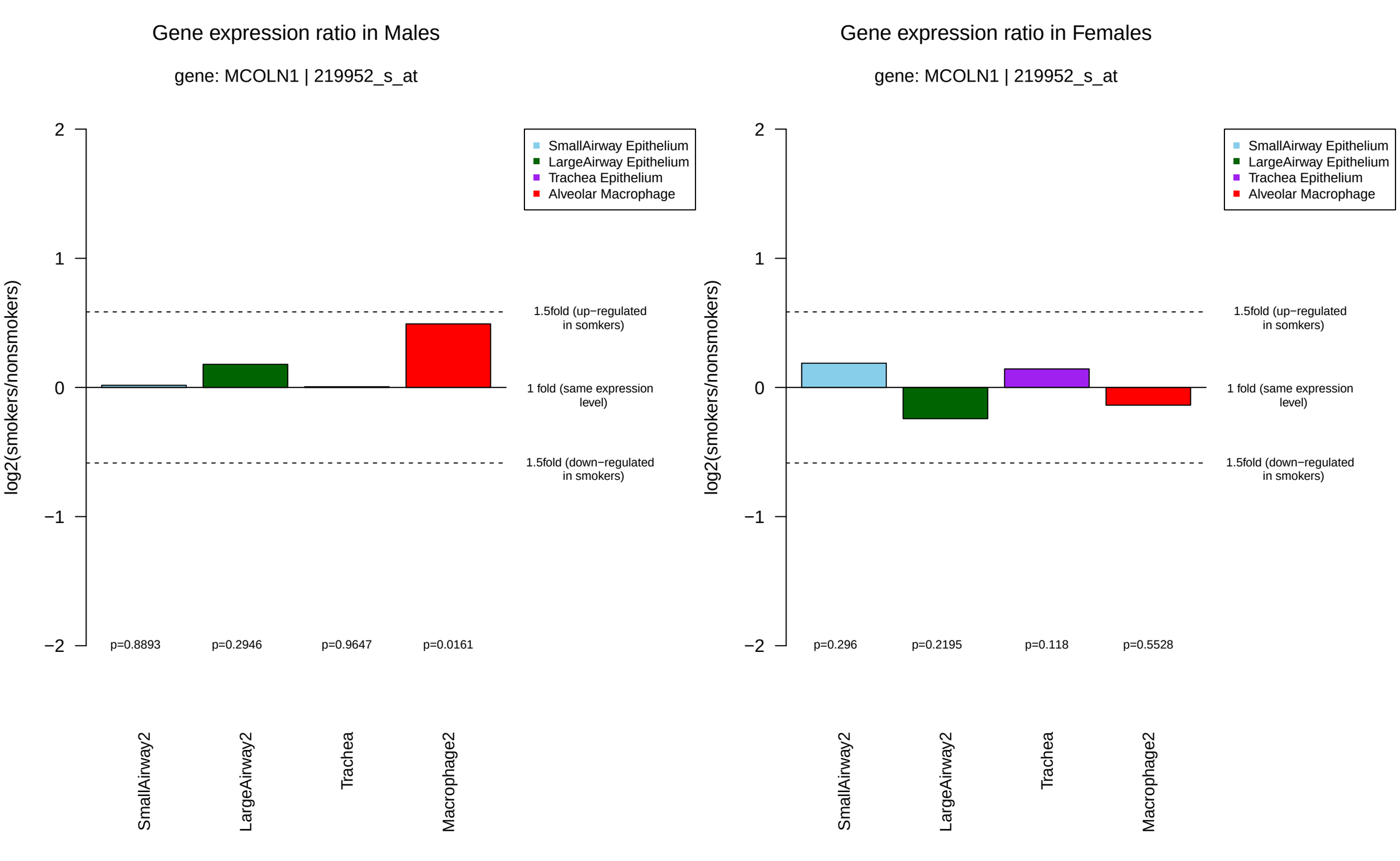
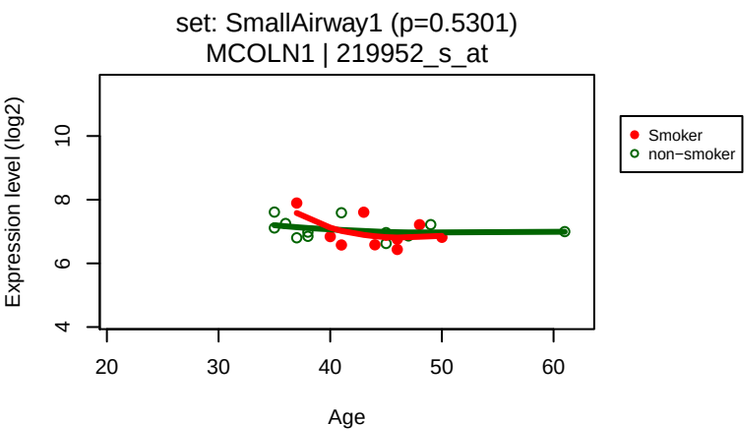
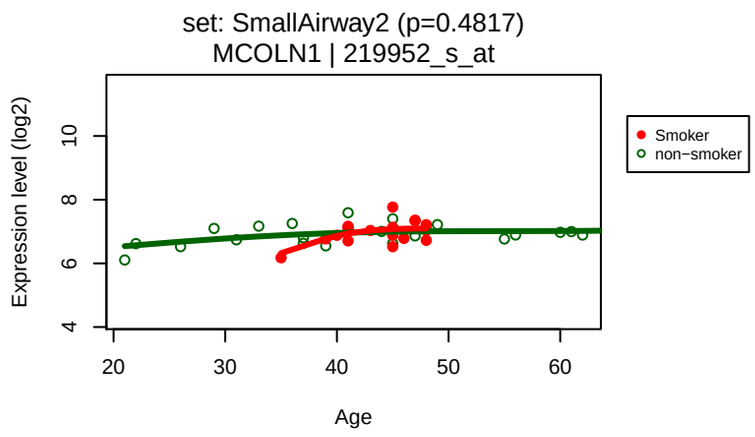
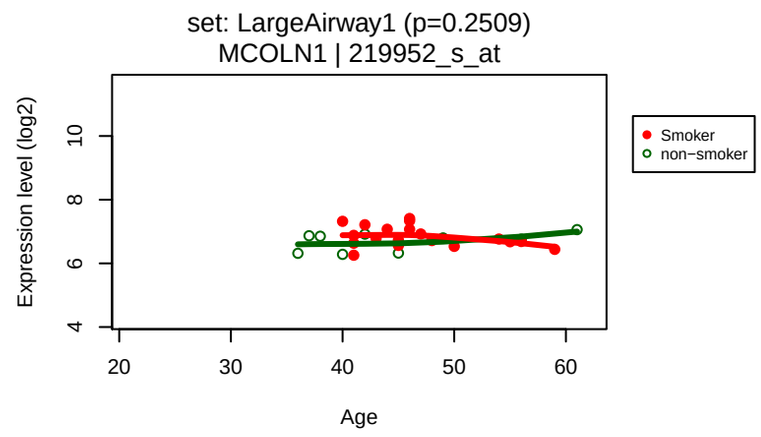

Gene expression of smokers and nonsmokers
gene: MCOLN1 | 219952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5301
p=0.4817
p=0.4284
p=0.2509
p=0.809
p=0.7167
p=0.0181
p=0.07
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

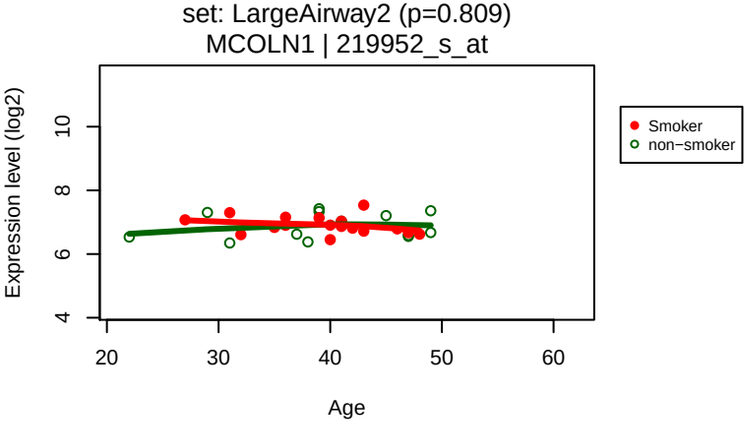

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN1 | 219952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5301
p=0.4817
p=0.4284
p=0.2509
p=0.809
p=0.7167
p=0.0181
p=0.07
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

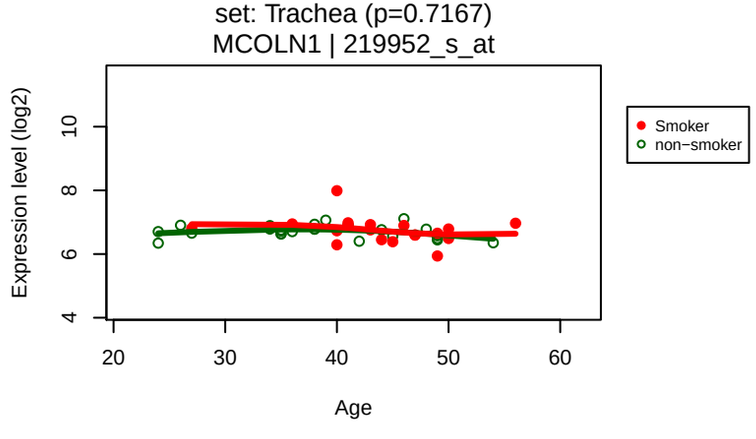

Gene expression ratio in Males
gene: MCOLN1 | 219952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8893
p=0.2946
p=0.9647
p=0.0161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN1 | 219952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.296
p=0.2195
p=0.118
p=0.5528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

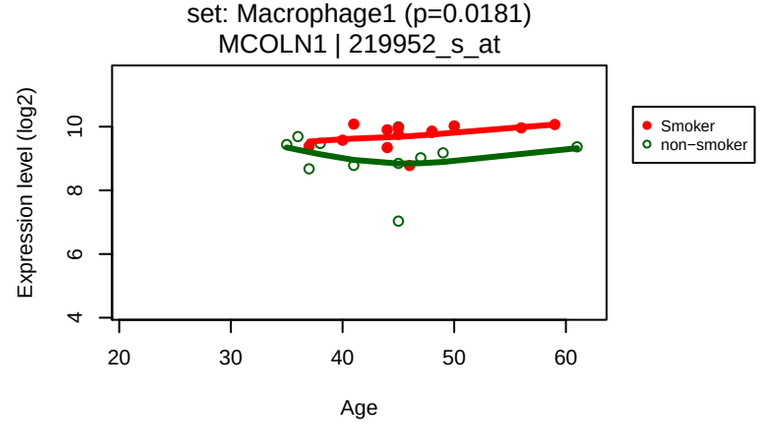

set: SmallAirway1 (p=0.5301)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker

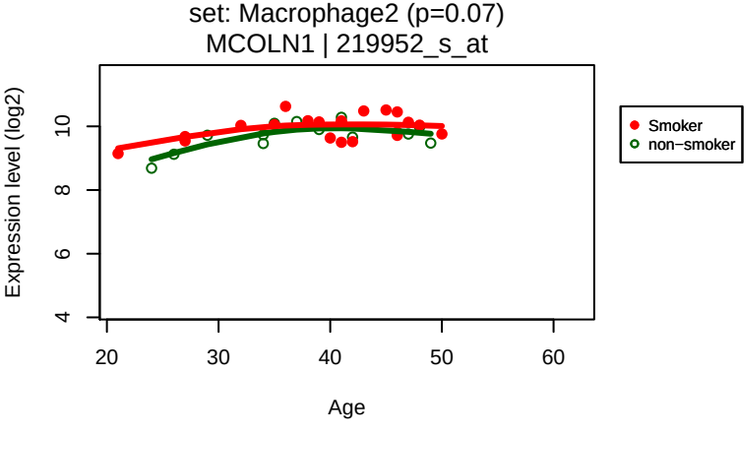

set: SmallAirway2 (p=0.4817)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker

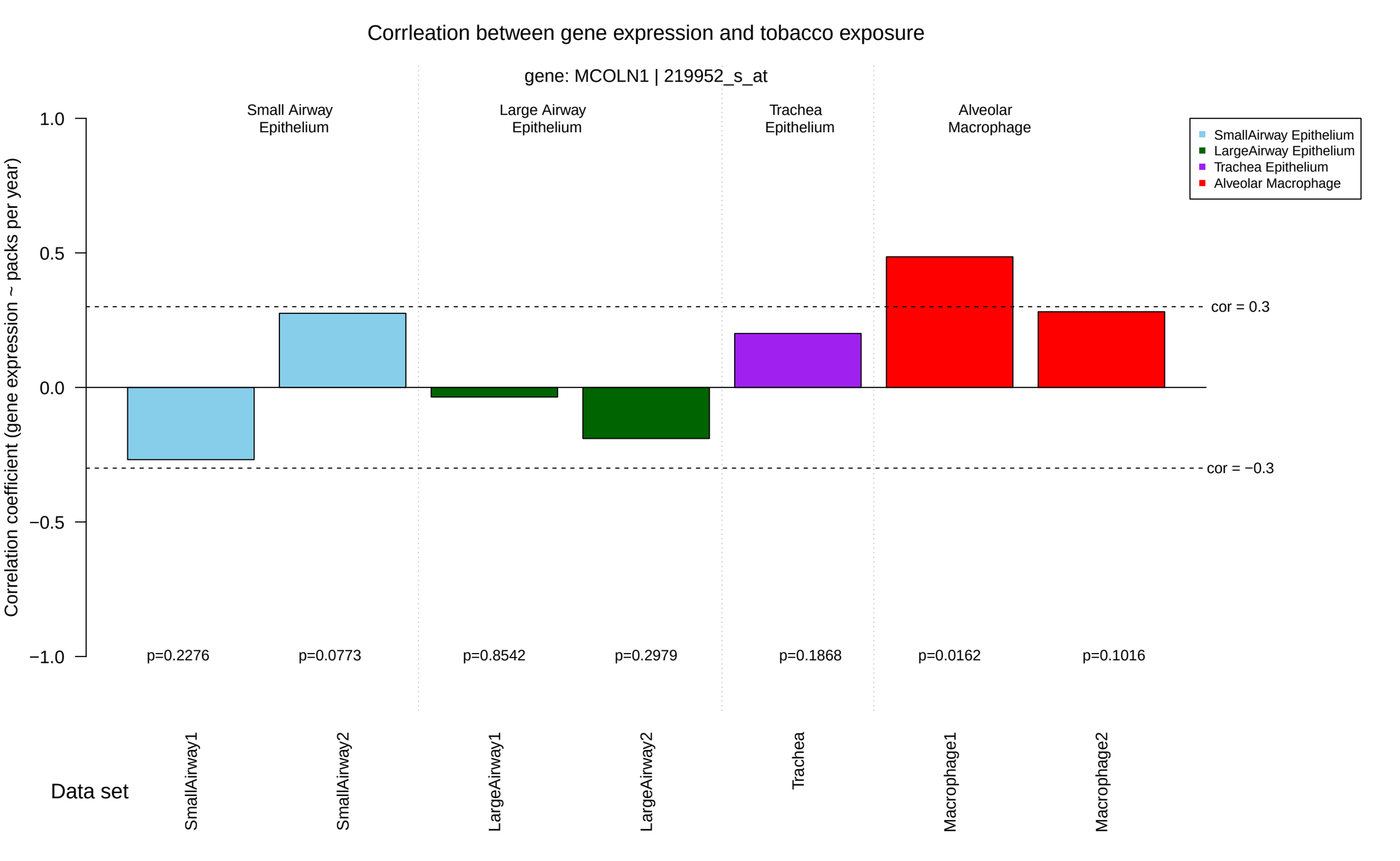

set: LargeAirway1 (p=0.2509)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.809)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7167)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0181)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.07)
MCOLN1 | 219952_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN1 | 219952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2276
p=0.0773
p=0.8542
p=0.2979
p=0.1868
p=0.0162
p=0.1016
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2